Algorithms Analysis Midterm 2023-2024

MSQ

1- Algorithm analysis is determining the amount of___resources necessary to execute it.

a) one b) two c) **three** d) four

2- Algorithm _____means how does the input/output relation match algorithm requirement?

a) clarity b) optimality c) simplicity d) correctness

3- _____essential approaches are used to analyze the algorithm.

a) Five b) Four c) Three d) Two

4- The _____order of growth rate means that when *N* doubles, runtime increases fourfold.

a) 2N b) N^2 c) *N2* d) 2^N

5- The insertion and removal in the ___has an O(1) in the best, average, and worst cases.

a) tree b) stack c) graph d) queue

True or False

1- Many problems are in a complexity class for which no practical algorithms are known. [True]

2- The semantics of an algorithm is its actual representation (body). [False]

3- The efficiency of algorithm determines its complexity. [False]

4- Analyzing the average case of an algorithm is the most difficult in practice. [True]

5- An algorithm is a procedure that maps one or more input(s) to one correct output. [True]

True or False

6- The approach to analyzing algorithms must be independent of four influences. [True]

7- Being "*O(n3)*" is a running time, or a function which gives the running time of algorithm. [False]

8- $f(n) \in O(g(n)) \equiv g(n) \in \Omega(f(n))$ [True]

9- Asymptotic growth rate is not always useful for analysis on fixed-size inputs. [True]

10- The bubble, insertion, merge and selection algorithms are incremental comparison sort. [False]

Essay

1 Consider the following algorithm:

- ***n = read input from user***
- ***pos = 0***
- ***neg = 0***
- ***i = 0***
- ***while i < n***
- ***number = read input from user***
- **if (number > 0)**
- **pos = pos + 1**
- **else if (number < 0)**
- **neg = neg + 1**
- **i = i + 1**
- ***write pos to user***
- ***write neg to user***

Complete Essay

1- Consider the following algorithm:

- *n = read input from user* ***1***
- *pos = 0* ***1***
- *neg = 0* ***1***
- *i = 0* ***1***
- *while i < n* ***1***
- *number = read input from user* ***1***
- if (number > 0) **1**
- pos = pos + **1**
- else if (number < 0)**OR**
- neg = neg + 1 **1**
- i = i + 1 **1**
- *write pos to user* ***1***
- *write neg to user* ***1***

n

Answer Essay

A. What is the computation time, *T(n)*, in terms of the input size *n*?

5n+6

B. What is the tight upper bound?

O(n)

C. Give an input sequence that satisfies the best case.

All positives (number>0)

D. Give an input sequence that satisfies the average case.

Positives, Zero and Negatives

E. Give an input sequence that satisfies the worst case.

All negatives (number<0)

Thank You